

# Topic 2

## Kinematics

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Revision Booklet

**This booklet covers:**

- Distance, Displacement, Speed, Velocity and Acceleration
- Graphs of Motion
- Equations of Uniform Acceleration (SUVAT)
- Free Fall and Measuring  $g$
- Projectile Motion

## Core Definitions

### Scalar and Vector Quantities

- **Distance** (scalar): the total length of path travelled, regardless of direction. Units: m.
- **Displacement**  $s$  (vector): the straight-line distance from start to finish *in a specified direction*. Units: m.
- **Speed** (scalar): distance travelled per unit time. Units:  $\text{m s}^{-1}$ .
- **Velocity**  $v$  (vector): rate of change of displacement. Units:  $\text{m s}^{-1}$ .
- **Acceleration**  $a$  (vector): rate of change of velocity. Units:  $\text{m s}^{-2}$ .

### Defining Equations

$$v = \frac{\Delta s}{\Delta t} \qquad a = \frac{\Delta v}{\Delta t}$$

$\Delta s$  = change in displacement (m)

$\Delta v$  = change in velocity ( $\text{m s}^{-1}$ )

$\Delta t$  = time interval (s)

### Distance vs Displacement

A car that travels 400 m north then 400 m south has a total **distance** of 800 m but a **displacement** of zero. Always identify whether the question asks for a scalar or vector quantity before calculating.

## Graphs of Motion

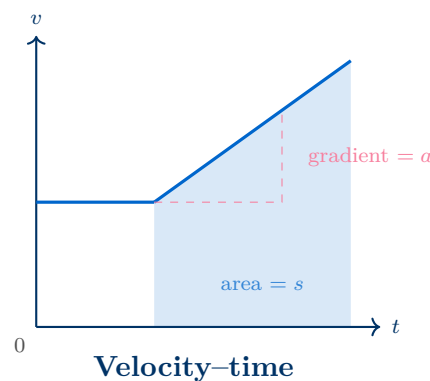
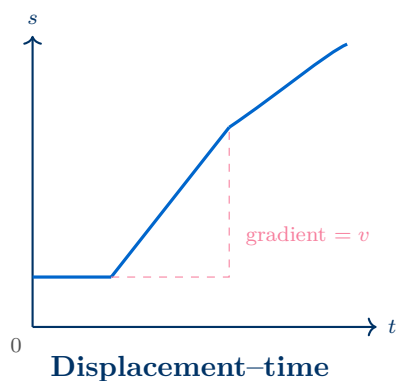
### Reading Motion Graphs

#### Displacement–time ( $s-t$ ) graph:

- Gradient = instantaneous velocity.
- Horizontal line  $\Rightarrow$  stationary.
- Straight line  $\Rightarrow$  constant (uniform) velocity.
- Curve  $\Rightarrow$  changing velocity (acceleration present).

#### Velocity–time ( $v-t$ ) graph:

- Gradient = instantaneous acceleration.
- Area under the graph = displacement.
- Horizontal line  $\Rightarrow$  constant velocity (zero acceleration).
- Straight line  $\Rightarrow$  uniform acceleration.



## Equations of Uniform Acceleration (SUVAT)

### Derivation

For **constant** acceleration, starting from the definitions of velocity and acceleration:

$$1. \text{ From } a = \Delta v / \Delta t: \quad a = (v - u) / t \quad \Rightarrow \quad \boxed{v = u + at}$$

$$2. \text{ Mean velocity} \times \text{time}: \quad s = \frac{u + v}{2} \times t \quad \Rightarrow \quad \boxed{s = \frac{1}{2}(u + v)t}$$

$$3. \text{ Substitute equation (1) into (2): } s = \frac{u + u + at}{2} t \quad \Rightarrow \quad \boxed{s = ut + \frac{1}{2}at^2}$$

$$4. \text{ Eliminate } t: \text{ from (1), } t = (v - u) / a; \text{ substitute into (2): } \Rightarrow \quad \boxed{v^2 = u^2 + 2as}$$

## The SUVAT Equations

| Equation                   | Variable not present |
|----------------------------|----------------------|
| $v = u + at$               | $s$                  |
| $s = ut + \frac{1}{2}at^2$ | $v$                  |
| $v^2 = u^2 + 2as$          | $t$                  |
| $s = \frac{1}{2}(u + v)t$  | $a$                  |

$s$  = displacement (m)       $u$  = initial velocity ( $\text{m s}^{-1}$ )  
 $v$  = final velocity ( $\text{m s}^{-1}$ )       $a$  = acceleration ( $\text{m s}^{-2}$ )       $t$  = time (s)

Valid only for **uniform** (constant) acceleration in a straight line.

## SUVAT Problem Strategy

1. Write down the five variables:  $s$ ,  $u$ ,  $v$ ,  $a$ ,  $t$ .
2. Fill in the known values; identify what you are finding.
3. Choose the equation that contains those four variables (three known + one unknown).
4. Solve; check units and sign conventions.

Positive direction must be defined clearly at the start — be consistent throughout.

Free Fall and Measuring  $g$ 

## Free Fall

**Free fall** is the motion of an object under gravity alone, with no air resistance. The object accelerates uniformly downwards at the **acceleration of free fall**:

$$g = 9.81 \text{ m s}^{-2} \quad (\text{near Earth's surface})$$

Free fall is uniform acceleration, so all four SUVAT equations apply with  $a = g$ .

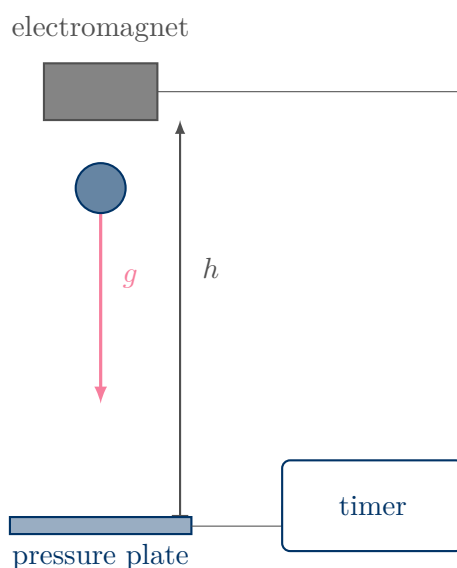
## Experiment: Determining $g$ Using a Falling Object

### Method — Electromagnetic Release and Trapdoor Timer

1. A steel ball is held by an electromagnet at a measured height  $h$  above a trapdoor switch.
2. The circuit is broken simultaneously: the ball is released and a millisecond timer starts.
3. When the ball hits the pressure plate the timer stops, giving the time of fall  $t$ .
4. Use  $h = \frac{1}{2}gt^2$  (with  $u = 0$ ) to find  $g$ :

$$g = \frac{2h}{t^2}$$

5. Repeat for several values of  $h$ ; plot  $h$  against  $t^2$  — the gradient equals  $g/2$ .



### Sources of Uncertainty and Improvements

- **Reaction time:** eliminated by electronic (not manual) timing.
- **Residual magnetism:** the ball may not release instantly — add a thin bit of tape between the ball and the magnet.
- **Air resistance:** use a dense, compact sphere to minimise drag.
- **Height measurement:** measure from the *bottom* of the ball to the pressure plate; use a ruler held vertically alongside.
- **Graph method:** plotting  $h$  vs  $t^2$  and finding the gradient  $= g/2$  averages over many measurements, reducing random error.

## Projectile Motion

### Projectile Motion

A **projectile** moves under gravity alone after launch. Its motion resolves into two **independent** components:

- **Horizontal:** uniform velocity (no force, so no acceleration).
- **Vertical:** uniform acceleration downwards at  $g = 9.81 \text{ m s}^{-2}$ .

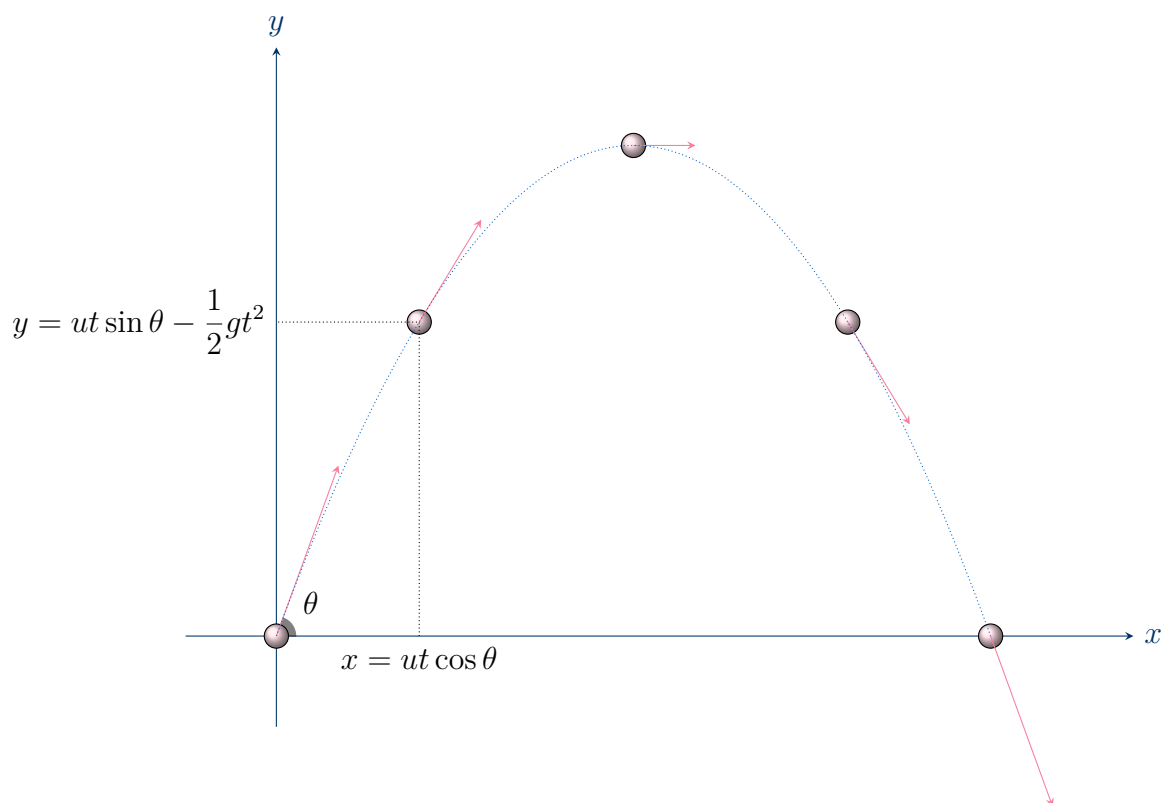
The two components share the same time  $t$  but are otherwise treated separately.

### Projectile Equations

For a projectile launched with speed  $u$  at an angle  $\theta$  to the horizontal :

|                          | Horizontal            | Vertical                               |
|--------------------------|-----------------------|----------------------------------------|
| Initial velocity         | $u_x = u \cos \theta$ | $u_y = u \sin \theta$                  |
| Acceleration             | 0                     | $g$ (downwards)                        |
| Velocity at time $t$     | $v_x = u \cos \theta$ | $v_y = u \sin \theta - gt$             |
| Displacement at time $t$ | $x = ut \cos \theta$  | $y = ut \sin \theta - \frac{1}{2}gt^2$ |

Resultant speed:  $v = \sqrt{v_x^2 + v_y^2}$       Angle from horizontal:  $\theta = \arctan\left(\frac{v_y}{v_x}\right)$



### Common Mistakes in Projectile Problems

- Never mix horizontal and vertical quantities in the same SUVAT equation.
- The horizontal velocity **does not change** (no air resistance assumed).
- Time is the same for both components — find  $t$  from the vertical motion first.
- For a horizontal launch:  $u_x = u$ ,  $u_y = 0$ .

## Worked Examples

### Example 1 — SUVAT: Braking Car

**Question:** A car travelling at  $30 \text{ m s}^{-1}$  brakes uniformly and stops in 4.5 s. Calculate (a) the deceleration and (b) the stopping distance.

#### Solution

**Known:**  $u = 30 \text{ m s}^{-1}$ ,  $v = 0$ ,  $t = 4.5 \text{ s}$

(a) Use  $v = u + at$ :

$$0 = 30 + a \times 4.5 \quad \Rightarrow \quad a = \frac{-30}{4.5} = -6.7 \text{ m s}^{-2}$$

(b) Use  $s = \frac{1}{2}(u + v)t$ :

$$s = \frac{1}{2}(30 + 0)(4.5) = 67.5 \text{ m}$$

### Example 2 — Free Fall

**Question:** A stone is dropped from rest off a cliff and hits the water 3.2 s later. Calculate (a) the height of the cliff and (b) the speed on impact.

#### Solution

**Known:**  $u = 0$ ,  $a = 9.81 \text{ m s}^{-2}$ ,  $t = 3.2 \text{ s}$

(a) Use  $s = ut + \frac{1}{2}at^2$ :

$$s = 0 + \frac{1}{2}(9.81)(3.2)^2 = \frac{1}{2} \times 9.81 \times 10.24 = 50.2 \text{ m}$$

(b) Use  $v = u + at$ :

$$v = 0 + 9.81 \times 3.2 = 31.4 \text{ m s}^{-1}$$

### Example 3 — Projectile Motion

**Question:** A ball is launched horizontally at  $15 \text{ m s}^{-1}$  from a platform 20 m above the ground. Calculate (a) the time of flight and (b) the horizontal range.

#### Solution

(a) **Vertical motion** ( $u_y = 0$ ,  $a = 9.81 \text{ m s}^{-2}$ ,  $s = 20 \text{ m}$ ):

$$20 = \frac{1}{2}(9.81)t^2 \quad \Rightarrow \quad t^2 = \frac{20}{4.905} \quad \Rightarrow \quad t = 2.02 \text{ s}$$

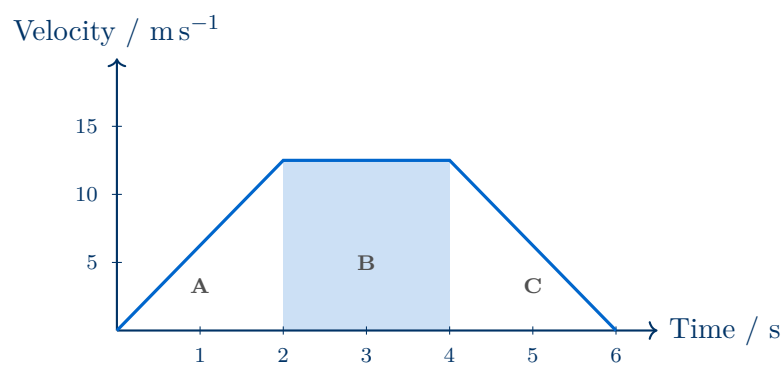
(b) **Horizontal motion** ( $u_x = 15 \text{ m s}^{-1}$ , constant):

$$x = u_x \times t = 15 \times 2.02 = 30.3 \text{ m}$$

**Practice Exam Questions****Section A — Short Answer Questions**

**Q1.** Distinguish between *distance* and *displacement*, and between *speed* and *velocity*.  
[4 marks]

**Q2.** The velocity–time graph below shows the motion of an object. Describe the motion in each labelled region and state what the shaded area represents.



[4 marks]

**Q3.** Define acceleration and state its SI unit.

[2 marks]

**Q4.** State the two conditions under which the SUVAT equations are valid.

[2 marks]

## Section B — Longer Structured Questions

**Q5.** A student drops a ball bearing from rest and measures the time  $t$  it takes to fall a height  $h$ . The results are shown in the table below.

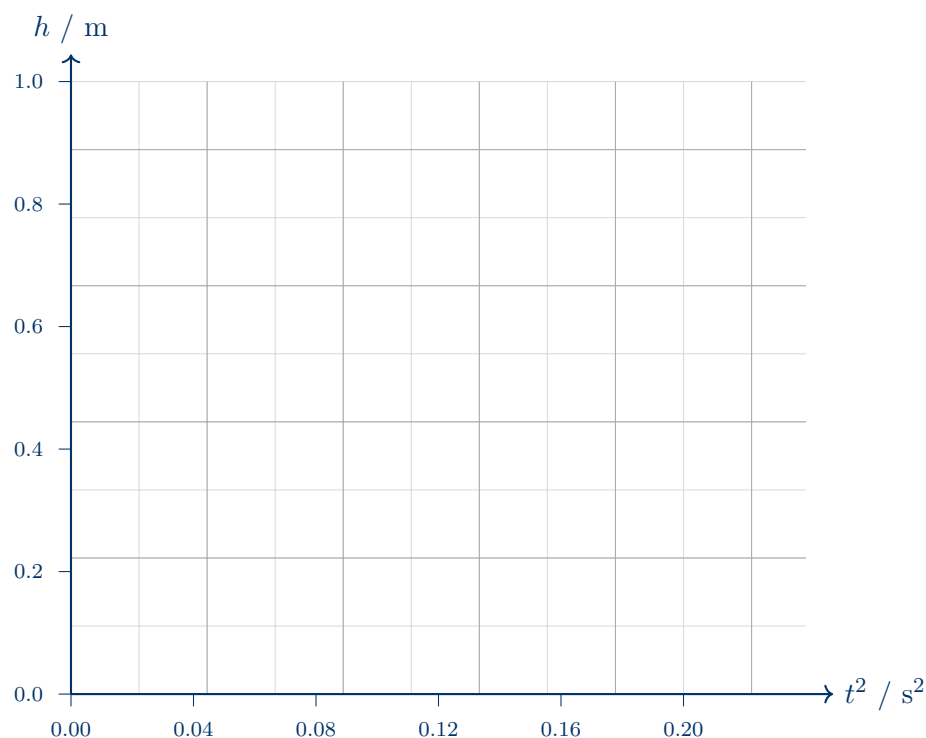
| Height $h$ / m | Time $t$ / s | $t^2$ / s <sup>2</sup> |
|----------------|--------------|------------------------|
| 0.20           | 0.202        |                        |
| 0.40           | 0.285        |                        |
| 0.60           | 0.350        |                        |
| 0.80           | 0.404        |                        |
| 1.00           | 0.451        |                        |

- (a) Complete the  $t^2$  column in the table above.

[1 mark]

- (b) Plot a graph of  $h$  (y-axis) against  $t^2$  (x-axis) on the grid below and draw a best-fit straight line.

[3 marks]



(c) Determine the gradient of your line and use it to calculate  $g$ .

*[3 marks]*

(d) Suggest one source of systematic error in this experiment and explain how it would affect the value of  $g$  obtained.

*[2 marks]*

**Q6.** A ball is kicked horizontally from the top of a vertical cliff at a speed of  $12 \text{ m s}^{-1}$ . The ball lands 35 m from the base of the cliff.

(a) Show that the time of flight is approximately 2.9 s.

*[2 marks]*

(b) Calculate the height of the cliff.

*[2 marks]*

(c) Calculate the speed and direction of the ball just before it hits the ground.

[3 marks]

## Mark Scheme and Answers

**Q1.** Distance is a scalar — the total length of path travelled [1]; displacement is a vector — the straight-line distance in a specified direction from start to finish [1]. Speed is the scalar rate of change of distance [1]; velocity is the vector rate of change of displacement [1].

**Q2.** Region A: uniform acceleration (straight line, positive gradient) [1]; Region B: constant velocity (horizontal line, zero acceleration) [1]; Region C: uniform deceleration to rest (straight line, negative gradient) [1]; shaded area = displacement during region B [1].

**Q3.** Acceleration is the rate of change of velocity [1]; SI unit:  $\text{m s}^{-2}$  [1].

**Q4.** Acceleration must be **uniform** (constant) [1]; motion must be in a **straight line** [1].

**Q5(a).**  $t^2$  values (3 s.f.): 0.0408, 0.0812, 0.1225, 0.1632, 0.2034  $\text{s}^2$  [1].

**Q5(b).** Points plotted correctly to within half a small square [1]; sensible scale used [1]; best-fit straight line through or close to the origin [1].

**Q5(c).** Gradient =  $\Delta h / \Delta t^2$  read from the line using a large triangle [1]; since  $h = \frac{1}{2}gt^2$ , gradient =  $g/2$  [1];  $g = 2 \times \text{gradient} \approx \mathbf{9.8 \text{ m s}^{-2}}$  [1].

**Q5(d).** Any valid systematic error, e.g. residual magnetism delays the ball's release, so the ball already has a small downward velocity when the timer starts [1]; this makes the measured  $t$  smaller than the true fall time, so  $t^2$  is underestimated, the gradient is too large, and the calculated value of  $g$  is **too high** [1].

**Q6(a).** Horizontal:  $x = u_x t$ , so  $t = 35/12 = 2.92 \text{ s} \approx 2.9 \text{ s}$  [2].

**Q6(b).**  $h = \frac{1}{2}gt^2 = \frac{1}{2} \times 9.81 \times (2.92)^2 = \mathbf{41.8 \text{ m}}$  [2].

**Q6(c).**  $v_y = gt = 9.81 \times 2.92 = 28.6 \text{ m s}^{-1}$  [1]; resultant speed =  $\sqrt{12^2 + 28.6^2} = \mathbf{31.0 \text{ m s}^{-1}}$  [1]; angle below horizontal =  $\arctan(28.6/12) = \mathbf{67.2^\circ}$  [1].

## Revision Checklist

Use this checklist to track your understanding. Tick each box when you are confident:

| Learning Objective                                                                                       | Confidence (1–3) |
|----------------------------------------------------------------------------------------------------------|------------------|
| <input type="checkbox"/> Define and distinguish distance, displacement, speed, velocity and acceleration |                  |
| <input type="checkbox"/> Determine velocity from the gradient of a displacement–time graph               |                  |
| <input type="checkbox"/> Determine acceleration from the gradient of a velocity–time graph               |                  |
| <input type="checkbox"/> Determine displacement from the area under a velocity–time graph                |                  |
| <input type="checkbox"/> Derive the four SUVAT equations from the definitions of $v$ and $a$             |                  |
| <input type="checkbox"/> Select and apply the correct SUVAT equation to solve problems                   |                  |
| <input type="checkbox"/> Describe the free-fall experiment and identify sources of uncertainty           |                  |
| <input type="checkbox"/> Use $h = \frac{1}{2}gt^2$ and a graph of $h$ vs $t^2$ to determine $g$          |                  |
| <input type="checkbox"/> Resolve projectile motion into independent horizontal and vertical components   |                  |
| <input type="checkbox"/> Calculate range, time of flight, and velocity components for a projectile       |                  |

*Key: 1 = Need more work    2 = Getting there    3 = Confident*

### Good luck with your revision!

Remember: always define a positive direction before starting a SUVAT problem, and keep horizontal and vertical components completely separate in projectile questions.